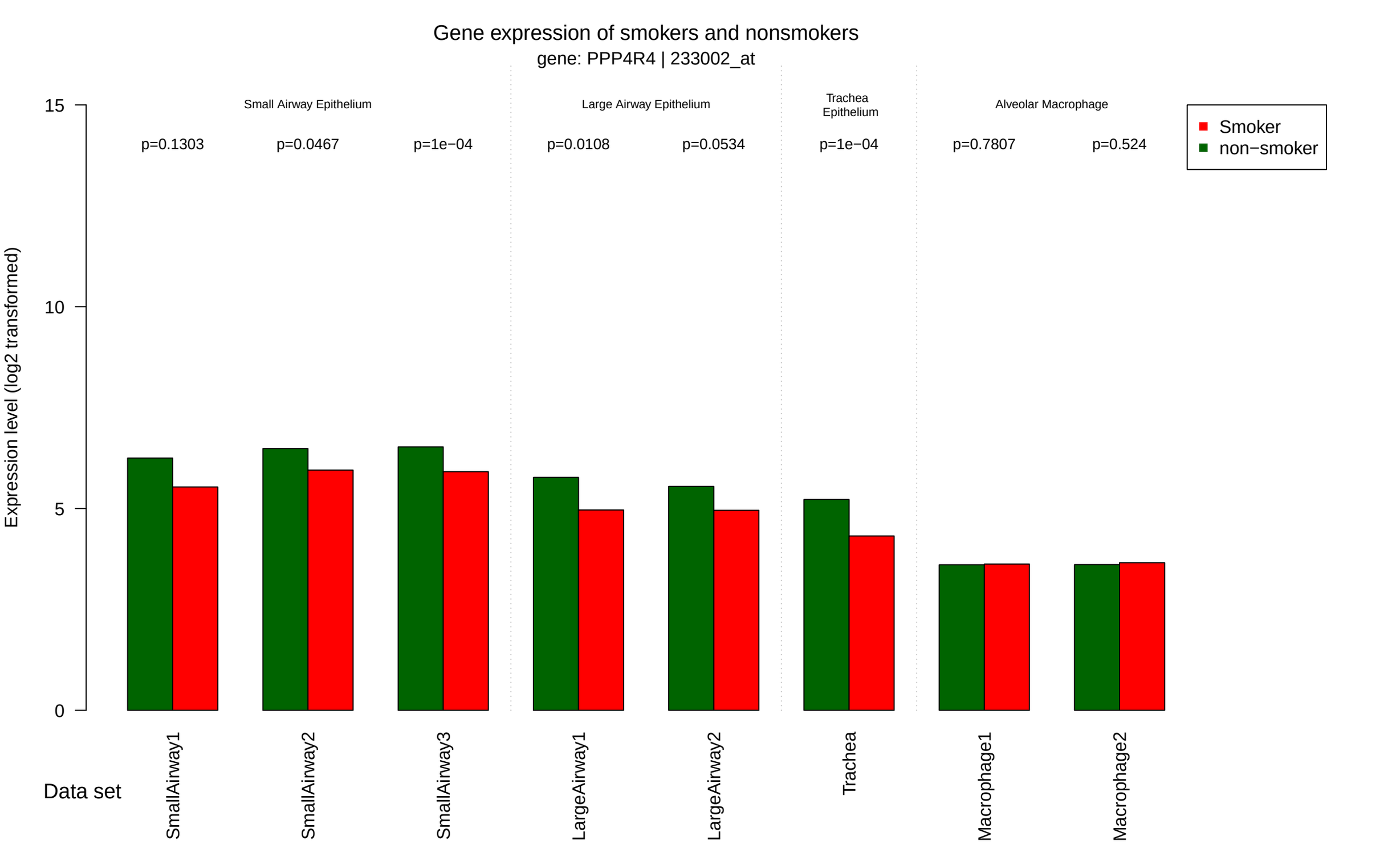
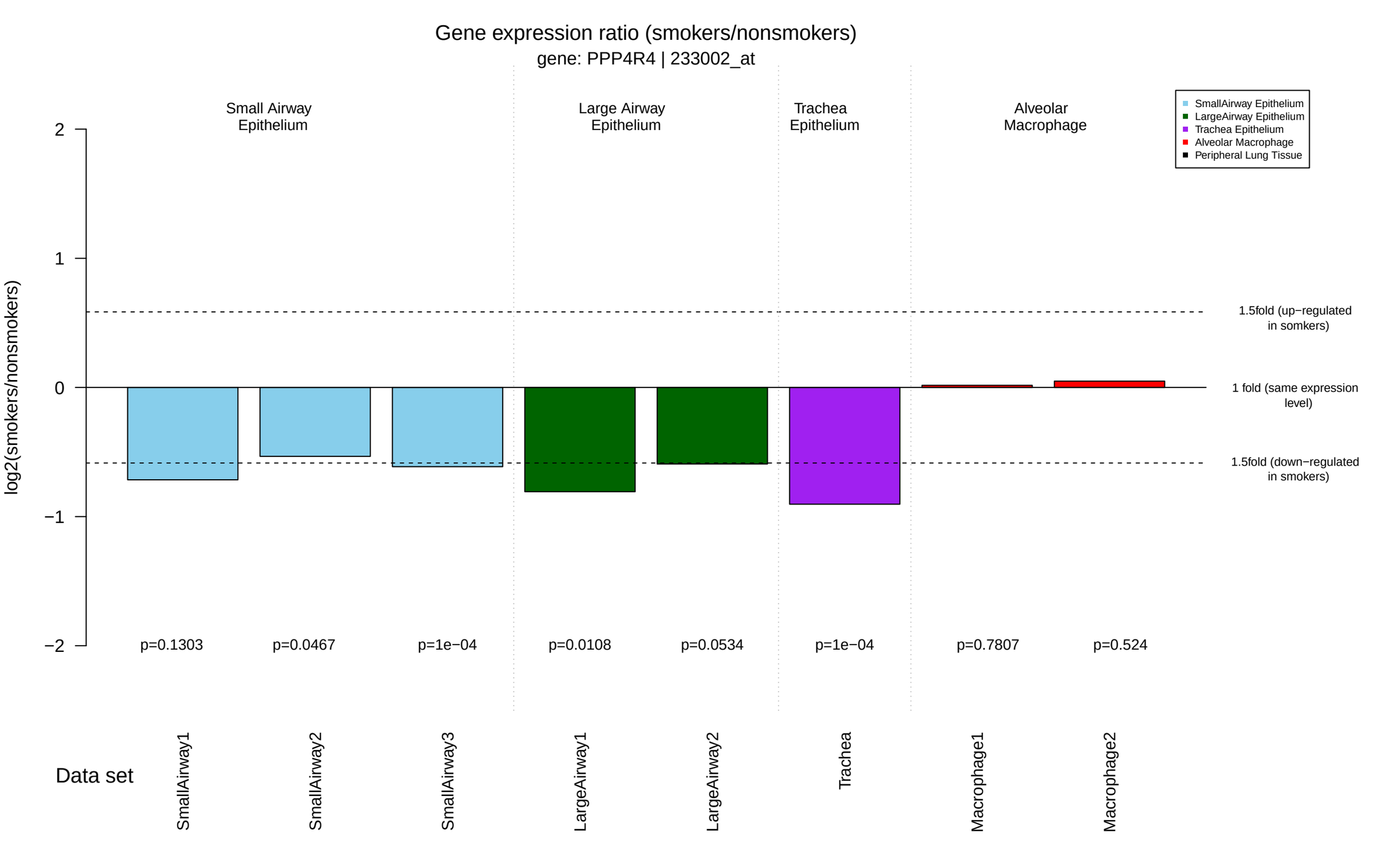
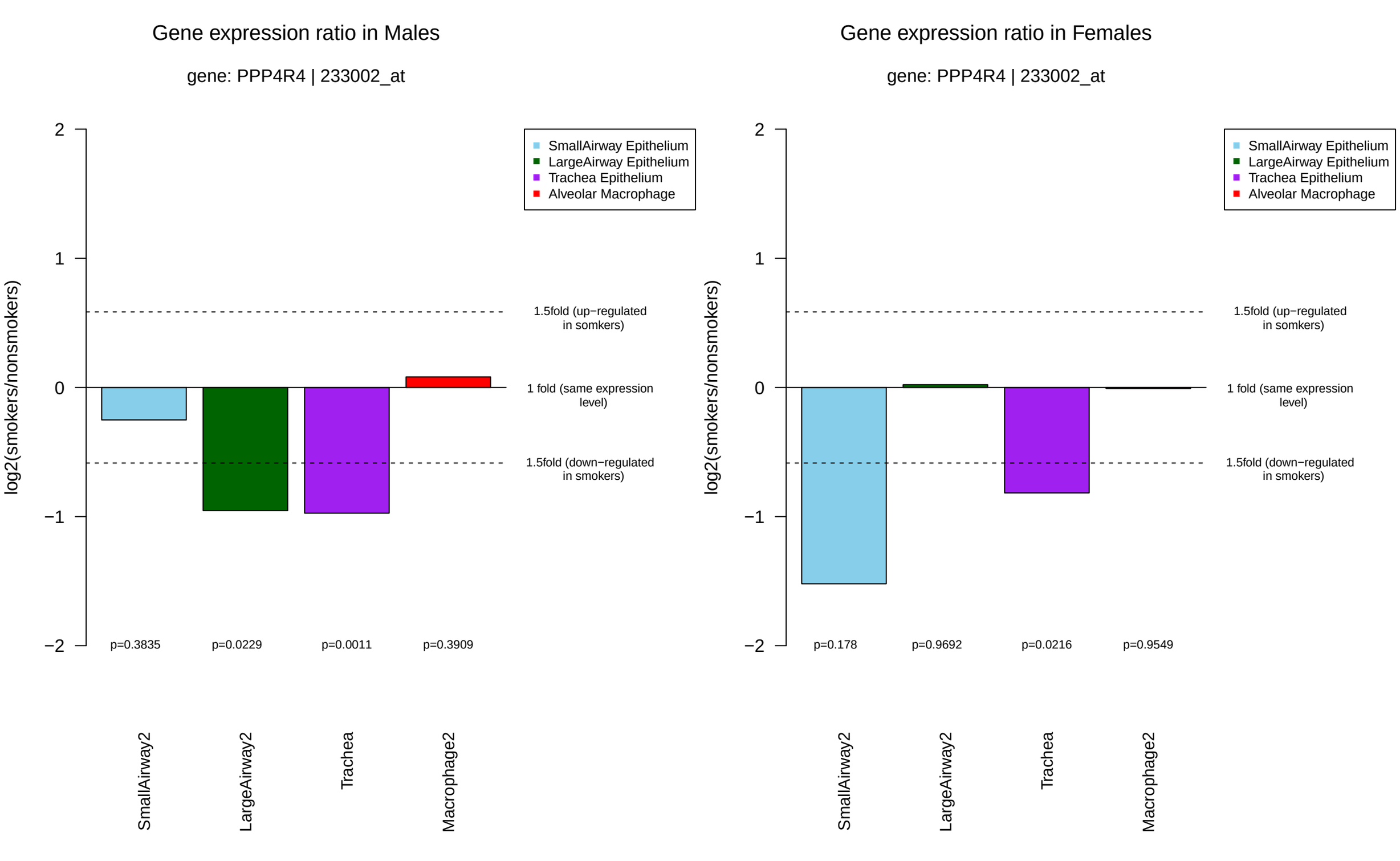
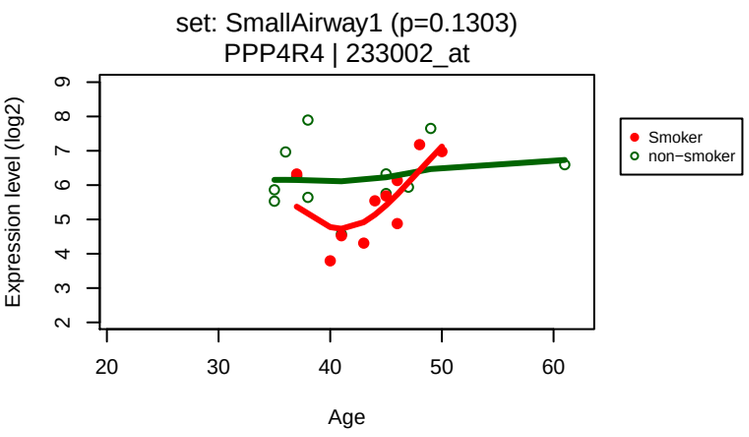
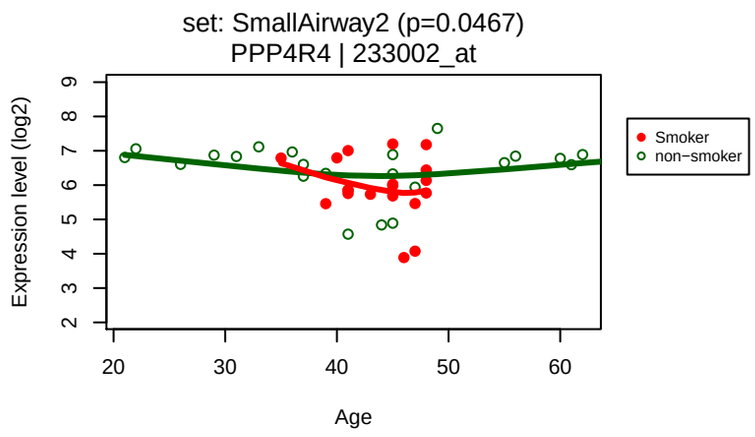
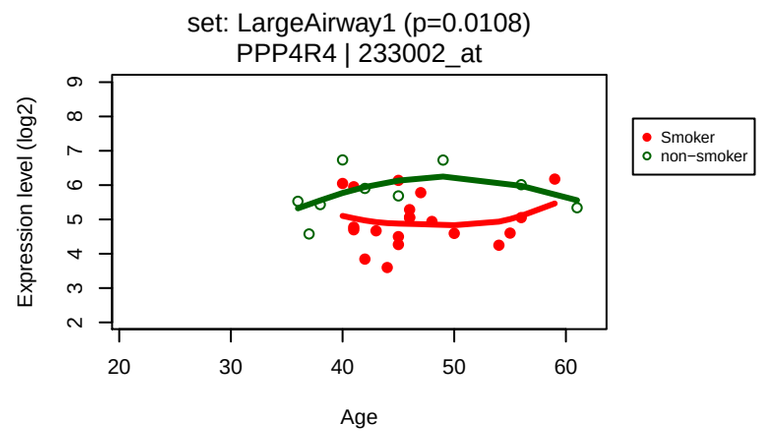

Gene expression of smokers and nonsmokers
gene: PPP4R4 | 233002_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1303
p=0.0467
p=1e−04
p=0.0108
p=0.0534
p=1e−04
p=0.7807
p=0.524
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

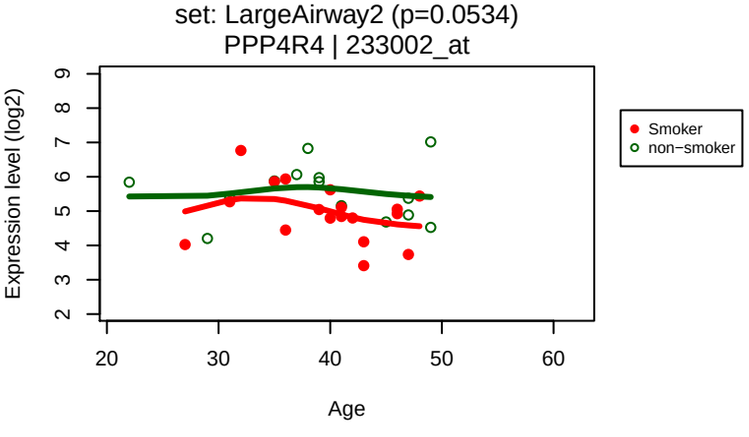

Gene expression ratio (smokers/nonsmokers)
gene: PPP4R4 | 233002_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1303
p=0.0467
p=1e−04
p=0.0108
p=0.0534
p=1e−04
p=0.7807
p=0.524
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

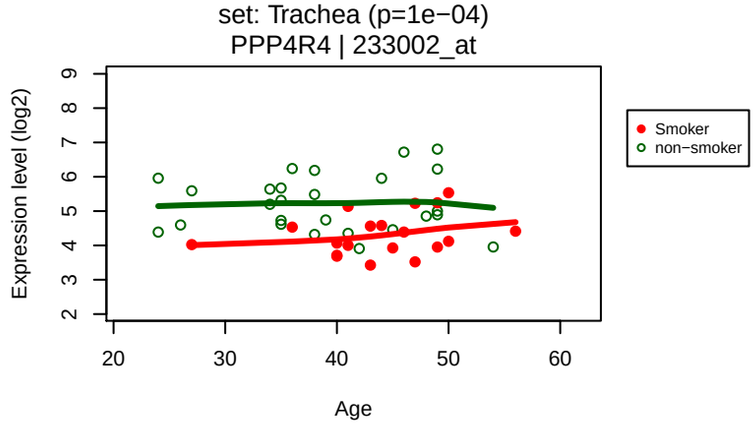

Gene expression ratio in Males
gene: PPP4R4 | 233002_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3835
p=0.0229
p=0.0011
p=0.3909
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPP4R4 | 233002_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.178
p=0.9692
p=0.0216
p=0.9549
SmallAirway2
LargeAirway2
Trachea
Macrophage2

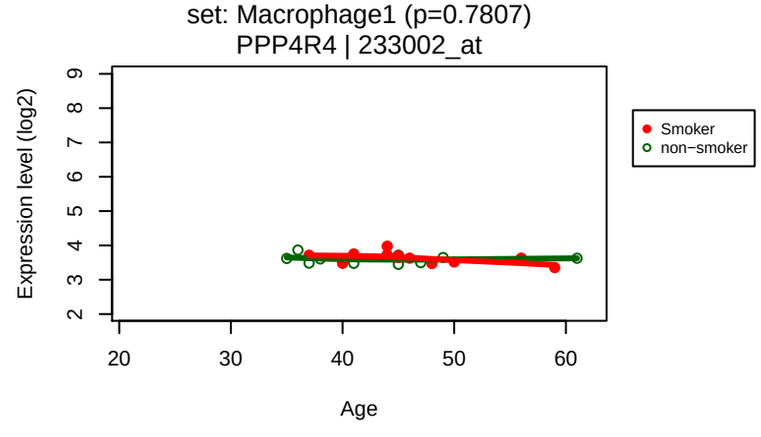

set: SmallAirway1 (p=0.1303)
PPP4R4 | 233002_at
Expression level (log2)
Age
Smoker
non−smoker

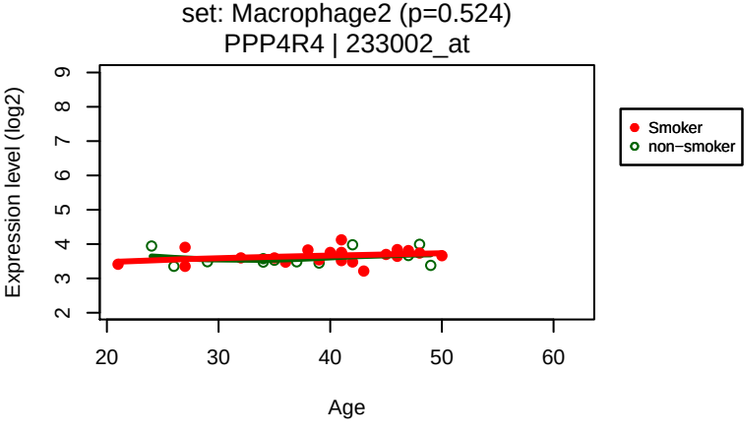

set: SmallAirway2 (p=0.0467)
PPP4R4 | 233002_at
Expression level (log2)
Age
Smoker
non−smoker

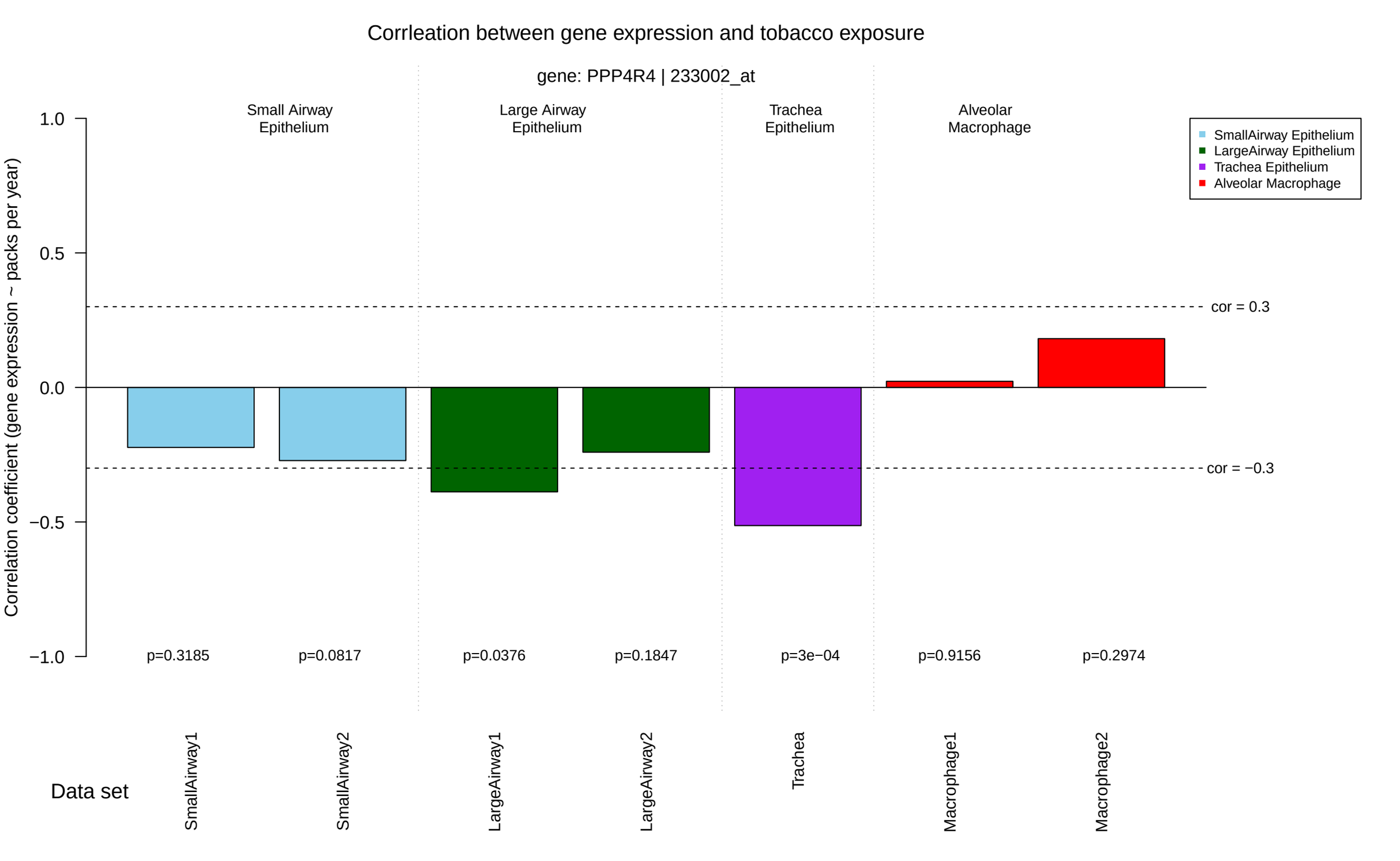

set: LargeAirway1 (p=0.0108)
PPP4R4 | 233002_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0534)
PPP4R4 | 233002_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
PPP4R4 | 233002_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7807)
PPP4R4 | 233002_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.524)
PPP4R4 | 233002_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPP4R4 | 233002_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3185
p=0.0817
p=0.0376
p=0.1847
p=3e−04
p=0.9156
p=0.2974
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2